Oesophagitis and Its Complications in Children: Strictures, Barrett's Oesophagus in At-Risk Children, and the Eosinophilic Differential.

A fourteen-year-old boy comes to clinic to be transferred to adult care. He was born with oesophageal atresia, the condition in which the upper and lower parts of the oesophagus fail to join, and he had the common form, with a fistula between the lower oesophagus and the trachea. He was repaired in the first days of life, had a few dilations of the join as a toddler, took acid suppression for his first two years, and has had no endoscopy since. He eats normally and says he has no heartburn. His mother asks whether he needs a camera test at all, given that he feels well. The honest answer is yes, and this chapter explains why.

The clinical question is this: when does reflux leave a scar? The main idea is that in an otherwise healthy child, reflux oesophagitis is common, usually shallow, and heals with treatment. The scarring complications, meaning strictures and a change in the type of cells lining the oesophagus, cluster in a small group of children whose anatomy, motility or lung disease makes reflux severe and lifelong. These are mainly children with repaired oesophageal atresia, children with severe neurological impairment, and children with cystic fibrosis. In them, symptoms are a poor guide, so surveillance should be scheduled rather than prompted by complaints. And every biopsy, in every child, should also look for the great mimic, eosinophilic oesophagitis.

The sources include the 2018 paediatric reflux guideline from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN and ESPGHAN, written by Rosen and colleagues; the 2016 joint guideline on gastrointestinal complications of oesophageal atresia by Krishnan and colleagues; the 2024 ESPGHAN guideline on eosinophilic oesophagitis by Amil-Dias and colleagues; the 2018 international AGREE consensus on eosinophilic oesophagitis by Dellon and colleagues; the 2022 Barrett's guideline of the American College of Gastroenterology, known as the ACG, by Shaheen and colleagues; and the 2014 guideline of the British Society of Gastroenterology. The main studies are named as we go.

Part one. From injury to scar or metaplasia.

Let us follow the injury step by step. The first step is exposure. Gastric acid, the digestive enzyme pepsin and sometimes bile reach the oesophagus, and they stay there too long when clearance is poor: when the oesophagus moves badly, when the child lies flat for long periods, or when part of the stomach has slid up through the diaphragm, which is called a hiatal hernia.

The second step is injury to the lining. The oesophagus is lined by squamous epithelium, flat layered cells like those of the skin. Under repeated attack, the basal layer of these cells thickens, which pathologists call basal cell hyperplasia, the spaces between cells widen, and eventually the surface breaks down into erosions and ulcers. Visible breaks in the lining are what endoscopists call erosive oesophagitis, and they grade it with the Los Angeles classification, from grade A, with small breaks shorter than five millimetres, to grade D, with breaks around at least three quarters of the circumference.

The third step is healing, and the oesophagus heals in one of two ways. If an ulcer is deep, it heals with collagen, and the scar contracts into a short narrowing just above the junction with the stomach: a peptic stricture. Alternatively, the damaged squamous lining is replaced by a different type of lining, made of column-shaped glandular cells. This change of one cell type into another is called metaplasia, and a columnar lining in the oesophagus is called Barrett's oesophagus. In children, the first columnar lining to appear is usually of the gastric type, resembling the lining of the upper stomach. Over years it may turn into intestinal metaplasia, which contains goblet cells like those of the intestine. It is intestinal metaplasia that carries a risk of dysplasia and adenocarcinoma, and that risk unfolds over decades.

This chain explains why complications are rare in most children: their exposure has been short. The exceptions are children who have had severe reflux since infancy. After oesophageal atresia repair, the oesophagus moves abnormally and the junction with the stomach is often pulled upwards, so reflux is frequent and clearance is poor. Children with severe neurological impairment lie flat, have spastic muscles, swallow poorly, and often have a hiatal hernia and scoliosis. Children with cystic fibrosis cough constantly and have overinflated lungs, which raises the pressure on the stomach. In these groups, the left-hand side of the chain is active for years.

Part two. How common is erosive oesophagitis, and how deep does it go?

How often erosive oesophagitis is found depends on who is examined. In a large American paediatric endoscopy database, reported by Gilger and colleagues in 2008, it was found in 12.4 percent of about seven thousand two hundred procedures, roughly one in eight. Among 402 neurologically normal children and young people who were scoped for reflux symptoms in Houston, reported by El-Serag and colleagues in 2002, it was found in 34.6 percent, about one in three. And in a single-centre review of 3,286 diagnostic endoscopies published by Edwards and colleagues in 2022, it was found in only 1.5 percent, but more than half of those children, 55 percent, were neurologically impaired.

Most paediatric oesophagitis is shallow. In a randomised trial of esomeprazole in children aged one to eleven, reported by Tolia and colleagues in 2010, 53 children had erosive

disease at the start: 32 had grade A, 19 grade B, and only one each had grades C and D. After eight weeks of once-daily esomeprazole at doses between 0.2 and 1.0 milligrams per kilogram, 40 of the 45 children who were scoped again had healed, which is 89 percent, with similar results across the doses. The practical message is that severe oesophagitis, grade C or D, is unusual in a child and should make you look for a reason, such as a hiatal hernia, atresia or neurological impairment. Adult practice adds one more step. The 2022 ACG guideline recommends a repeat endoscopy eight to twelve weeks after acid suppression in a patient with oesophagitis, because healing can reveal Barrett's lining that the inflammation was hiding; in the data it cites, Barrett's was found on repeat endoscopy in 17.4 percent after grade C or D oesophagitis and in 9.4 percent after grade A or B.

What about Barrett's in healthy children? It is rare. In the El-Serag series, Barrett's was suspected at endoscopy in 11 of 402 children, 2.7 percent, but none had intestinal metaplasia on biopsy. In a prospective study of 840 children without neurological or oesophageal abnormalities, reported by Nguyen and colleagues in 2011, Barrett's was suspected in 1.43 percent and confirmed in a single child, 0.12 percent. A columnar-looking tongue in a healthy child is usually not intestinal metaplasia, and true Barrett's in such a child should prompt a search for an underlying cause.

Part three. The neurologically impaired child.

Children who cannot describe pain present differently. In the Edwards series, 26 neurologically impaired children with erosive oesophagitis were compared with 21 other children; most of the impaired children had cerebral palsy, and 77 percent had documented reflux. Haematemesis, vomiting blood, was the reason for endoscopy in 65.4 percent of the impaired children compared with 23.8 percent of the others. Any sign of blood loss was present in 84.6 percent compared with 28.6 percent. Severe oesophagitis, grades three and four on the Hetzel and Dent scale, was found in 79 percent compared with 59 percent. A faecal occult blood test was positive in 91.7 percent of the impaired children with oesophagitis, against 33.3 percent of the others. And no impaired child presented with abdominal pain. Afterwards, 42.3 percent of the impaired children went on to fundoplication, an antireflux operation that wraps the top of the stomach around the lower oesophagus, compared with 4.8 percent of the others. This was a small retrospective series, so read these figures as a strong signal rather than as rates. The bedside lesson is that in a child with neurological impairment, iron-deficiency anaemia, blood in vomit, food refusal or a change in behaviour may be the only signs of severe oesophagitis.

Part four. Strictures.

Not every stricture is peptic, and the location and shape point to the cause. A peptic stricture is short and distal, at the junction with the stomach, usually with active oesophagitis above it, and it is typical of the neurologically impaired child, the child with a hiatal hernia, and the child with atresia. An anastomotic stricture sits at the site of the atresia repair, in the middle of the oesophagus. An eosinophilic stricture tends to be long, with a narrow calibre throughout, often with rings and furrows, and it is classically discovered when an adolescent has food stuck in the oesophagus. Other causes include caustic ingestion, the blistering skin disease epidermolysis bullosa, and injury from tablets. The first rule is never to dilate blind to the cause. Take biopsies above and at the stricture, because active reflux oesophagitis needs acid suppression, eosinophilic disease needs its own treatment, and a dilation done into an untreated inflammatory stricture will simply have to be repeated.

Anastomotic strictures after atresia repair are common. In a Dutch national multicentre study by Vergouwe and colleagues, published in the Archives of Disease in Childhood in 2019, 436 children with an end-to-end repair were followed. Fifty-eight percent developed a stricture, needing a median of three dilations. Seven percent, 32 children, developed a refractory stricture, needing a median of ten dilations and up to thirty-four. Three factors predicted a refractory course in the multivariable analysis. A first dilation needed within twenty-eight days of the repair carried an odds ratio of 15.9, meaning that the odds of a refractory stricture were about sixteen times higher. Isolated atresia without a fistula, which is usually a long-gap atresia repaired under tension, carried an odds ratio of 5.7. An anastomotic leak, which occurred in 13 percent of repairs, carried an odds ratio of 5.0. The ESPGHAN consensus definition of a refractory stricture is a narrowing without endoscopic inflammation that still causes difficulty swallowing after at least five dilations at intervals of no more than four weeks. Adult endoscopists use a slightly different definition, from Kochman and colleagues: failure to reach a diameter of fourteen millimetres after five sessions at two-week intervals.

Three further points on atresia strictures are worth knowing. Giving acid suppression routinely to prevent an anastomotic stricture does not work; acid suppression treats proven reflux, not the anastomosis. An international randomised trial called STEPS-EA is testing steroid injections into the stricture to prevent refractory strictures. And the 2016 guideline accepts fundoplication for recurrent strictures when reflux remains uncontrolled despite maximal acid suppression, after full evaluation.

The management of a symptomatic stricture follows a logical order. First define it, with a contrast study to show its length and number, and an endoscopy with biopsies. Second, treat the cause: a proton pump inhibitor for erosive disease, and for eosinophilic disease dietary elimination, a proton pump inhibitor or a swallowed topical steroid. Third, dilate with a balloon or a bougie in small steps, and only if the child has symptoms; the 2016

guideline finds no support for routine dilation of an asymptomatic anastomosis. Fourth, for a refractory stricture, consider steroid injection into the stricture, topical mitomycin C, a temporary stent or incision of the scar, knowing that the evidence for each is weak. Fifth, consider surgery: fundoplication when reflux drives recurrence, and resection only as a last resort. For eosinophilic strictures, the 2024 ESPGHAN guideline allows a course of systemic steroids before dilation, to reduce inflammation and make dilation safer and more durable.

Part five. What counts as Barrett's oesophagus?

Before quoting any rate of Barrett's in children, ask which definition was used, because the societies disagree. The 2022 ACG guideline requires columnar lining extending at least one centimetre above the top of the gastric folds, and it requires intestinal metaplasia on biopsy. The 2014 British Society of Gastroenterology guideline also uses the one-centimetre threshold, but does not require intestinal metaplasia, so gastric-type metaplasia counts. Paediatric series, especially in atresia, usually count any columnar lining, which is mostly of the gastric type. And the 2018 paediatric reflux guideline does not define Barrett's at all; it simply lists it among the complications an endoscopy can find. By the ACG definition, most of what is reported as Barrett's in children would not qualify. The practical response is to biopsy by the Seattle protocol, which means biopsies from all four quadrants every two centimetres along the columnar segment plus any visible lesion, and to report gastric and intestinal metaplasia separately. For adults with Barrett's without dysplasia, the ACG recommends surveillance endoscopy every five years for segments shorter than three centimetres, and every three years for longer segments.

Part six. Barrett's oesophagus after atresia repair.

For children with atresia, Barrett's is a lifelong question. The best overview is a systematic review by Tullie and colleagues in BJS Open in 2021, which gathered 6,282 patients followed for more than two years after repair. Barrett's was reported in 5.0 percent of all patients, 317 people, but in 12.8 percent of the 1,727 patients who were actively examined in screening or surveillance programmes. In other words, the more you look, the more you find, and Barrett's is undercounted wherever nobody scopes. Of the 317 cases, 227 were gastric metaplasia and 54 were intestinal metaplasia. The mean age at diagnosis was 13.8 years, but in screened patients intestinal metaplasia was found at a mean age of 38.5 years. So metaplasia starts in childhood, but becomes intestinal in adulthood.

The same review found thirteen oesophageal cancers. The surprise is that nine were squamous cell carcinomas and only four were adenocarcinomas, at a mean age of about forty years and a range of twenty to forty-seven years. This means that surveillance in

adult survivors of atresia must examine the whole squamous oesophagus, including the scar of the anastomosis and the dilated upper pouch, and not only the junction with the stomach.

How common is metaplasia by adolescence? Schneider and colleagues, in a prospective French multicentre study published in the Annals of Surgery in 2016, scoped 120 adolescents aged fifteen to nineteen with repaired atresia, at a mean age of sixteen and a half. Seventy percent had been treated for reflux in infancy, and 41 percent had had antireflux surgery. At evaluation, 41 percent had reflux symptoms. On biopsy, 67 percent had oesophagitis and 43 percent had Barrett's: fifty with gastric metaplasia and one with intestinal metaplasia. Crucially, symptoms did not predict Barrett's. The risk factors were atresia without a fistula, repeated antireflux surgery, previous dilations and histological oesophagitis. No abnormality was found at the anastomosis itself in this young group.

This is why the 2016 joint guideline treats endoscopy with biopsies as mandatory for monitoring reflux in atresia, and suggests at least three routine endoscopies in childhood: one after stopping acid suppression, one before the age of ten, and one at transition to adult care. A single-centre Italian study by Maestri and colleagues in 2022 followed 75 children through this schedule. Histological oesophagitis was found in 72 percent, and eosinophilic oesophagitis in 12 percent, nine children, far above the rate in the general population. Four of the 35 children who had no symptoms at all had eosinophilic oesophagitis. After transition there is no agreed adult interval; some experts suggest regular screening from the late teens or early twenties.

Cystic fibrosis is the other group to remember. In a study from Columbia University by Knotts and colleagues, comparing adults with cystic fibrosis with matched controls, cystic fibrosis carried about a threefold risk of Barrett's oesophagus or oesophageal cancer, and both appeared at younger ages. A long segment of columnar lining in a teenager with cystic fibrosis therefore deserves the same attention as in an adult, with referral to a Barrett's surveillance programme at transition.

Part seven. Reflux or eosinophilic oesophagitis?

Eosinophilic oesophagitis is a chronic allergic inflammation of the oesophagus, defined by symptoms of oesophageal dysfunction together with at least fifteen eosinophils per high-power field on biopsy. Several features help to separate it from reflux, although none is perfect on its own. The typical patient with reflux is an infant with regurgitation or an older child with heartburn, while the typical patient with eosinophilic oesophagitis is an atopic boy with difficulty swallowing or food impaction. At endoscopy, reflux causes distal erosions graded by the Los Angeles system, whereas eosinophilic oesophagitis causes rings, longitudinal furrows, white exudates and a narrow calibre, which are scored together as the EREFS score. In reflux, eosinophils are few and concentrated in the distal

oesophagus, with a gradient towards the junction; in eosinophilic oesophagitis they are numerous, at fifteen or more per high-power field, and present in both the proximal and the distal oesophagus, often patchily, which is why at least six biopsies from two levels are needed. Strictures in reflux are short and distal, while eosinophilic strictures are long or of small calibre, and the mucosa may tear at dilation.

The most important change of recent years concerns proton pump inhibitors. Children and adults with oesophageal eosinophilia who respond to a proton pump inhibitor were once labelled as having a separate condition. The 2018 AGREE consensus abolished that category: about half of patients with eosinophilic oesophagitis respond to a proton pump inhibitor, so a response does not exclude it, and a proton pump inhibitor is now one of its first-line treatments. The 2024 ESPGHAN guideline takes the same position.

The two diseases also overlap, especially after atresia. In the Maestri cohort, 12 percent of children with atresia had eosinophilic oesophagitis, and it was particularly frequent in the types without a distal fistula, three of seven children, compared with 9 percent in the common types. Reflux may damage the barrier and let food antigens in, and eosinophilic inflammation may stiffen the wall and impair clearance, increasing reflux contact. The 2016 guideline advises managing eosinophilic oesophagitis in atresia exactly as in other children. The practical consequence is to biopsy every stricture for eosinophils, and to re-biopsy on treatment before attributing persistent symptoms to reflux.

Part eight. What the guidelines say.

Comparing the paediatric and adult documents shows where they agree and where children fall into gaps. On the role of endoscopy they are aligned: the paediatric guideline uses it to assess complications and exclude eosinophilic disease rather than to diagnose reflux, and adult documents treat severe erosive oesophagitis, Barrett's and strictures as proof of reflux. On checking healing after severe oesophagitis there is a gap: the ACG advises repeat endoscopy at eight to twelve weeks, and the paediatric guideline does not address it. On the definition of Barrett's they differ, as we have seen. On surveillance in atresia the difference is material: the paediatric schedule ends at transition, and adult guidelines have no atresia-specific schedule, only general Barrett's intervals. On refractory strictures the definitions differ, four-week versus two-week intervals and a symptom-based versus a diameter-based endpoint. On eosinophilia that responds to a proton pump inhibitor, the paediatric and adult documents now agree that it is eosinophilic oesophagitis. The material gap, then, is that children with atresia graduate into adult services with a known risk of metaplasia, a risk of squamous cancer, and no agreed adult surveillance plan.

Part nine. Traps, and what to do differently in clinic.

Several reasoning errors are common. The first is to say that a child with atresia has no symptoms and so needs no endoscopy; in the French adolescents, symptoms did not predict Barrett's, so scope on the schedule. The second is to equate a salmon-coloured tongue with Barrett's; a hiatal hernia and an irregular junction mislead the eye, so biopsy and name the type of metaplasia. The third is to conclude that a response to a proton pump inhibitor proves reflux; eosinophilic oesophagitis responds too, and only the biopsy decides. The fourth is to dilate and see; untreated inflammation brings the stricture back, and an early need for dilation after atresia repair predicts a refractory course. And one more: looking only at the junction in an adult survivor of atresia, when most cancers after repair were squamous.

Return now to the fourteen-year-old boy at transition. He should have an endoscopy now, with biopsies by protocol, inspecting the whole oesophagus, including the anastomosis, and with biopsies at two levels for eosinophils; his results should be handed to an adult team with a plan for lifelong follow-up. The same logic resolves other common scenarios. A nine-year-old with heartburn and grade B oesophagitis needs eight weeks of a proton pump inhibitor, and no routine repeat endoscopy if symptoms resolve. A teenager with cerebral palsy and iron-deficiency anaemia, without complaints, needs an endoscopy, because severe oesophagitis is likely, followed by acid suppression and attention to feeding position and any hiatal hernia. A three-year-old after atresia repair with food sticking and a stricture at the anastomosis needs biopsies above and at the stricture, treatment of reflux or eosinophilic disease, and then dilation. A child whose stricture recurs for the third time in six weeks despite a proton pump inhibitor, with reflux documented on pH-impedance monitoring, enters the refractory pathway, with discussion of fundoplication or steroid injection. And an adolescent with difficulty swallowing, rings and furrows, who improved on a proton pump inhibitor, has eosinophilic oesophagitis until the biopsies say otherwise.

So, tomorrow in clinic, carry these points. Erosive oesophagitis is found in about one in eight paediatric endoscopies, is mostly grade A or B, and heals in about nine of ten children after eight weeks of a proton pump inhibitor. Barrett's in a healthy child is rare, confirmed in one of 840 children. Neurologically impaired children present with blood rather than pain. After atresia repair, more than half develop a stricture and about one in fourteen develops a refractory stricture, predicted by early dilation, a long gap and an anastomotic leak. Barrett's after atresia is reported in 5 percent overall and in about 13 percent of those actively scoped, and in 43 percent of French adolescents, mostly of the gastric type, without symptoms to warn you. Cancers after atresia are rare, appear around the age of forty, and are mostly squamous. Follow the three scheduled endoscopies in childhood, hand over a surveillance plan at transition, biopsy every stricture for eosinophils, and remember that a response to a proton pump inhibitor does not exclude

eosinophilic oesophagitis.